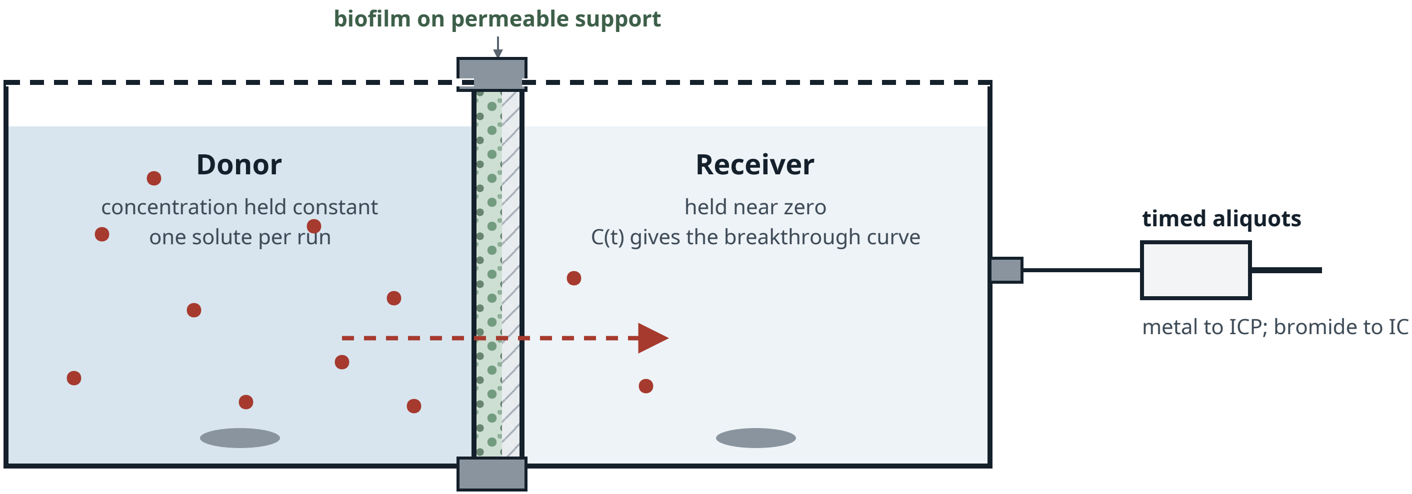
Phase 2 — two-chamber diffusion cell

Biofilm grown on a permeable support; transport read as receiver breakthrough



Both chambers stirred — unstirred boundary layers otherwise read as film resistance.

Paired blank: bare support

Identical cell and support, no biofilm, run alongside every measurement. Gives the support's own resistance and its wall sorption. Without it the result is film plus support, not film.

Edge seal checked separately: a leak reads as fast transport.

Open choice: the tracer

Bromide is size- and charge-excluded from fine porosity. In Hanford sediments tritiated water resolved intragranular pore volume where bromide did not; an anionic EPS matrix is the same trap.

Tritiated water carries its own approval burden. A question for Dr. Deng.

What one run yields

Breakthrough lag gives a lumped transport parameter, not a diffusivity alone: sorption retards it. Separating the two needs the partition coefficient from the Phase 1 suspended isotherm.

That ordering is why Phase 1 comes first.

What this number is not

The film-scale effective diffusivity measured here is not the reactor-scale dispersion coefficient that currently carries the same name in radial_parms (1×10^{-3} cm^2 s^{-1}, about 43× water's
It is the quantity slab_parms carries as D_eff (1×10^{-5} cm^2 s^{-1}) — the one default in that file below water self-diffusivity.